

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 2: Great Circles, Rhumb Lines & Convergence

Syllabus Source: CAE Oxford (pp. 43-84) | Pass Target: $\geq 90\%$ AviationExam

1. GREAT CIRCLES VS RHUMB LINES

Navigators navigate using two fundamental geometric curves on the Earth's sphere:

- Great Circle (Orthodrome): The intersection of the sphere with a plane passing through the CENTER of the Earth. **SHORTEST DISTANCE** between two points. It crosses successive meridians at constantly **CHANGING ANGLES**!
- Rhumb Line (Loxodrome): A line on the Earth's surface that crosses all meridians at the **SAME CONSTANT ANGLE**. Easy to fly (constant heading), but physically **LONGER** than a Great Circle!
- Relative Path: A Great Circle always curves **TOWARDS THE POLE** relative to the Rhumb Line connecting the same two points (curves North in N Hemisphere, curves South in S Hemisphere).

2. EARTH CONVERGENCE & CONVERSION ANGLE FORMULAS

Because meridians converge at the poles, the true direction of a Great Circle changes continuously along its track:

- Earth Convergence Formula: $\text{Earth Convergence (degrees)} = \text{Change of Longitude (dlong)} \times \sin(\text{Mean Latitude})$
- Conversion Angle Formula: $\text{Conversion Angle (CA)} = 0.5 \times \text{Earth Convergence} = 0.5 \times \text{dlong} \times \sin(\text{Mean Latitude})$
- Great Circle to Rhumb Line Relationship: $\text{Rhumb Line Track} = \text{Great Circle Track} \pm \text{Conversion Angle}$. In Northern Hemisphere: Great Circle Track from West to East starts **HIGHER** than Rhumb Line Track and finishes **LOWER**!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Great Circle Initial & Final Track

Problem: A flight links A (50°N, 020°W) and B (50°N, 040°E). The Rhumb Line track is 090°. Find the initial Great Circle track at A and final track at B. Step 1: Calculate change of longitude (Chlong) = 20°W + 40°E = 60°. Step 2: Earth Convergence = Chlong $\times$ sin(Mean Latitude) = 60° $\times$ sin(50°) = 60 $\times$ 0.766 = 46°. Step 3: Conversion Angle (CA) = 0.5 $\times$ Convergence = 0.5 $\times$ 46° = 23°. Step 4: Northern Hemisphere Great Circle curves towards the North Pole: - Initial Track at A = Rhumb Line - CA = 090° - 23° = 067°. - Midpoint Track = 090°. - Final Track at B = Rhumb Line + CA = 090° + 23° = 113°!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Great Circle Initial Track and Convergence

SCENARIO (Classic EASA Exam Question): An oceanic flight is planned between two waypoints in the Northern Hemisphere: - Point A: 50°N 010°W - Point B: 50°N 030°W - The Rhumb Line Track between A and B is 270° True (due West along 50°N parallel) QUESTION: What is the Earth Convergence, Conversion Angle, and INITIAL Great Circle Track from Point A to Point B? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Change of Longitude (dlong) and Mean Latitude: - dlong = 030°W - 010°W = 20°. - Mean Latitude = 50°N (both points are at 50°N). Step 2: Calculate Earth Convergence: - Formula: Earth Convergence = dlong x sin(Mean Latitude) - Convergence = 20° x sin(50°) = 20° x 0.766 = 15.32° (~15.3°). Step 3: Calculate Conversion Angle (CA): - Formula: Conversion Angle = 0.5 x Convergence - CA = 0.5 x 15.32° = 7.66° (~7.7°). Step 4: Determine Initial Great Circle Track from Point A: - Flying Westbound in Northern Hemisphere: The Great Circle curves towards the North Pole (curves to the right of the Rhumb Line)! - Therefore, the initial track points further North (towards 360°/clockwise): - Initial GC Track = Rhumb Line Track + CA = 270° + 7.7° = 277.7° True! - (And at Point B, the final GC track will be 270° - 7.7° = 262.3° True!). FINAL ANSWER: Earth Convergence = 15.3°; Conversion Angle = 7.7°; Initial Great Circle Track = 277.7° True.

[!] EXAM TRAP (AVIATIONEXAM) - The Convergence Zero at Equator Trap

- Earth Convergence = dlong x sin(lat). - At the EQUATOR (lat 0°), sin(0) = 0. Earth Convergence is ZERO! (Meridians are parallel). - At the POLES (lat 90°), sin(90) = 1. Earth Convergence equals dlong!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the Earth Convergence between longitudes 015°W and 045°W at a mean latitude of 30°N?**

- [A]: 30°
- [B]: 15° (dlong = 30°, Convergence = 30° x sin(30°) = 30° x 0.5 = 15°)
- [C]: 7.5°
- [D]: Zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. dlong = 45° - 15° = 30°. sin(30°) = 0.5. Earth Convergence = 30° x 0.5 = 15°.

Q2: In the Northern Hemisphere, how does a GREAT CIRCLE track curve relative to the RHUMB LINE connecting the two points?

- [A]: Curves towards the South Pole
- [B]: CURVES TOWARDS THE NORTH POLE (lies North of the Rhumb Line)
- [C]: It is perfectly straight
- [D]: Curves randomly

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. On a Mercator chart, Great Circles always bow towards the nearer geographic pole relative to the straight rhumb line.

Q3: What is the relationship between Conversion Angle and Earth Convergence?

- [A]: Conversion Angle = 2 x Convergence

- [B]: CONVERSION ANGLE = HALF OF EARTH CONVERGENCE ($CA = 0.5 \times \text{Convergence}$)
- [C]: They are identical
- [D]: Conversion Angle = $\sin(\text{lat})$

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Conversion Angle is defined as exactly half the Earth Convergence between two positions.